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RESEARCH ARTICLE

Deep learning-based super-resolution reconstruction and improved YOLOv9 for efficient benthos detection: a case study at Lake Hamana, Japan

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Abstract

The development of remote sensing and object detection technologies has advanced benthos surveys. However, challenges remain in accuracy and cost-efficiency due to environmental interference. A practical method combining drone-based image acquisition and deep learning techniques for benthos monitoring is presented. Field experiments objecting hermit crabs were conducted at Lake Hamana using drones at altitudes of 2 m, 5 m and 10 m. Super-resolution reconstruction (SRR) was applied to enhance image quality, followed by small-object detection using the self-built V9-BENTHOS. With a magnification factor × 4, Residual Dense Network (RDN) achieved optimal SRR performance (PSNR: 38.15 dB, SSIM: 88.51%) and V9-BENTHOS reached a mean average precision of 95.5%. The effects of SRR algorithms and magnification factors on hermit crab detection were discussed. This case study provides a new approach to support benthos ecological monitoring.

Introduction

Hermit crabs, named for their dependence on empty mollusc shells, are widely distributed across various marine ecosystems from intertidal zones to deep-sea environments (Greenshields et al., 2021). As a key component of marine ecosystems, hermit crabs contribute to organic detritus recycling and act as nutrient intermediaries in the marine food web (Hazlett, 1981). Furthermore, their behavioural ecology and adaptive strategies provide insights beyond assessing the health of the marine environment (Sanda et al., 2018). Owing to their sensitivity to environmental changes, widespread distribution and functional role within food chains, hermit crabs are considered effective indicators for evaluating the stability of marine ecosystems. For example, their reliance on empty snail shells makes them highly sensitive to changes in habitat structure, while their key role as detritivores in food webs allows their population size and health to directly reflect environmental stresses such as pollution and climate change.

Traditional hermit crab ecological surveys rely primarily on destructive methods, such as mark-recapture and biological sampling (Gauci et al., 2020; Ito, 2007; Makoto et al., 2019). While these methods can provide quantitative analysis, they often disrupt local population dynamics, are labour-intensive and have limited coverage. In contrast, the recent development of non-invasive remote sensing technologies, particularly the integration of imaging methods such as sonar and LiDAR, has expanded the

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scope and efficiency of benthic environmental monitoring (Alicia et al., 2023; Moniruzzaman et al., 2017; Solé et al., 2023). However, these technologies often lack species-specific recognition capabilities, necessitating the integration of methods with object recognition capabilities, such as UAV-based imagery (Chen et al., 2025; Zhao et al., 2024).

As an alternative to divers, Remotely Operated Vehicles (ROVs) and Autonomous Underwater Vehicles (AUVs) are increasingly being used because they offer greater safety and spatial coverage (Alicia et al., 2023; Kutser et al., 2020). However, their application is constrained by limited power capacity and rising maintenance costs caused by biofouling and corrosion during prolonged underwater deployments (Li et al., 2023). However, in shallow coastal areas, remote sensing methods such as satellite and drone imaging can provide long-term benthic biodiversity assessments (Xu et al., 2021). While satellite imagery provides large-scale information, its spatial resolution is insufficient to reveal centimetre-scale organisms, and its high cost limits its current ecological application to local scales (Fu et al., 2022).

In contrast, drone-based surveys offer greater flexibility, reproducibility and cost-effectiveness, making them particularly suitable for medium-scale ecological studies. Equipped with a variety of optical sensors, drones can generate high-resolution imagery and detect small benthic organisms, including hermit crabs, sea cucumbers and shellfish. Studies on the *Pagurus minutus* in particular have demonstrated its ecological vulnerability and importance as a species of key conservation concern (Henmi et al., 2014), highlighting the need for reliable species-specific monitoring. This makes drones potentially useful for species-specific ecological monitoring, bridging the gap between large-scale mapping and individual-level observations.

For processing the large amount of data, Support Vector Machines (SVM) were initially used to extract information from benthos images (Lutzenkirchen et al., 2024). Artificial Intelligence (AI) and Computer Vision (CV) have further enhanced image analysis capabilities to efficiently identify benthic organisms in large scenarios. Convolutional Neural Networks (CNNs), integrated on AUVs, have been used for object detection in underwater images to facilitate rapid environmental analysis (Liu et al., 2021; Piechaud et al., 2019). However, the global field of view algorithm only provides limited detection speed in species with simple structures such as clams and anemones (Fincham et al., 2020; Fu et al., 2022; Wang et al., 2026). For small-scale objects such as hermit crabs, the You Only Look Once (YOLO) series has significantly improved detection accuracy and speed through iterative modular (Zhao, Chen, et al., 2025; Zhao, Xu, et al., 2025). Now, it has been widely used in underwater image analysis, from highly manoeuvrable species such as fish to stationary organisms such as corals, from large objects such as seaweed to small organisms such as crabs and shrimps (Lyu et al., 2023; Wang, Sun, et al., 2024; Zhang et al., 2022; Zhou et al., 2024). YOLO also performs in the remote sensing images, especially in drone surveys (Li et al., 2025; You et al., 2025; Zhao et al., 2024). Taking advantage of the high manoeuvrability of drones, YOLO can detect small objects in real time and at high speed, improving survey efficiency.

For drone-based survey, national regulations and no-fly zone provisions impose restrictions on minimum flight altitudes, which limit the obtainable image resolution, resulting in a decrease in training data quality and affecting detection accuracy (Chuang & Sung, 2020). In addition, motion blur and ghosting effects lead to partial loss of information, making it hard to detect centimetre-level samples with high accuracy (Chuang & Sung, 2020). Traditional image enhancement techniques, such as filtering and histogram equalization, can only be supplemented based on existing pixel information (Wu et al., 2022). Super-resolution reconstruction (SRR) solves the key data limitation problem by enhancing high-altitude remote sensing images to a resolution comparable to ultra-low-altitude data. Interpolation-based SRR methods have simple structures and fast processing speeds, unable to handle sharp transitions and often produce aliasing and blocking effects, since simply filling pixel gaps cannot fully restore high-frequency details (Park et al., 2003). Deep learning-based SRR achieves full-frame resolution improvement with a relative computational cost. Since the introduction of the Super-Resolution Convolutional Neural Network (SRCNN) (Ward et al., 2017), various advanced architectures, including Enhanced Deep Super-Resolution network (EDSR) (Lim et al., 2017), Residual Channel Attention Network (RCAN) (Lin et al., 2022) and Super-Resolution Feedback Network (SRFBN) (Li et al., 2019), have been widely used. Although some studies have explored in underwater benthos detection, their application in remote sensing imagery, especially for centimetre-scale objects such as hermit crabs, remains a blank. Besides, high-resolution image acquisition usually requires expensive cameras, while SRR can enhance images taken with standard, cost-effective cameras, significantly reducing research costs while maintaining repeatability (Liu et al., 2021; Xiang et al., 2022; Zhao, Shao, et al., 2025).

Lake Hamana, located in Hamamatsu City, Shizuoka, Japan, serves as a representative habitat for hermit crabs. As a lagoon, it exhibits characteristics of both freshwater and saltwater ecosystems, providing a breeding ground for various hermit crab species. Research on hermit crabs

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in Lake Hamana dates to studies on the subspecies *Pagurus minutus* (Henmi et al., 2014). Unlike other similar species, *P. minutus* primarily inhabits snail shells and has a body length of only 1–2 cm, making population surveys particularly challenging. With increasing environmental pressures, the Japan Benthos Biological Association reported that between 1997 and 2012, 10 out of 17 hermit crab species in Lake Hamana had entered an extinction or quasi-extinction crisis, while the remaining 7 species lacked sufficient data for effective assessment. Therefore, developing efficient investigation methods is crucial for species conservation and maintaining ecological stability.

This study proposes a method for surveying hermit crab populations by integrating SRR and improved object detection algorithms, and conducts a case study in Lake Hamana. Specifically, low-altitude images of hermit crabs were taken at three nominal heights of 2 m, 5 m and 10 m in the shallow-water area of Lake Hamana using a consumer-grade drone to form datasets. Secondly, multiple SRR algorithms were tested to improve image resolution. Finally, the V9-BETHONS algorithm was developed for centimetre-level object detection.

The following structure of this paper is as follows: Section 2 illustrates the methodology, including the algorithm architecture and evaluation metrics for SRR and V9-BETHONS. Section 3 presents the reconstruction results of different SRR algorithms and evaluates the segmentation performance of V9-BETHONS from multiple perspectives. Section 4 discusses 2 technical advancements related to SRR and object detection. Finally, Section 5 concludes the study.

The training specification has been attached in the supplemental items.

Materials and Methods

In this study, a comprehensive ecological survey method integrating SRR was developed to efficiently collect hermit crab images and perform object detection, as illustrated in Fig. 2b. As shown in Fig. 2a, drones were deployed to capture aerial imagery of hermit crabs in Lake Hamana, generating original 4 K video (3840 × 2160 pixels, referred to as the original dataset) at 30 frames per second (fps). Frames were then extracted from the videos to construct a high-resolution (HR) dataset. To create low-resolution (LR) inputs, HR images were downsampled using bicubic interpolation. This method ensures consistent scene content between the low-resolution and high-resolution datasets, avoiding differences in lighting, perspective or motion that might arise when using the original high-altitude imagery. This ensures that the reconstructed images are identical in depicting the same spot, while also enabling fair comparison between datasets. Finally, the proposed V9-BENTHOS algorithm was applied for hermit crab detection. Further details will be illustrated in Sections 2.1–2.3.

Drone-based survey of hermit crabs

As shown in Fig. 1, the experiment was conducted in August 2023 at the southeastern shoal of Lake Hamana (34°44′N, 137°34′E), a key intersection for ocean current exchange between the inner lake and the Sea of Japan and a primary habitat of hermit crabs. During the experiment, a DJI Phantom 4 drone was deployed for aerial imaging, operated by two trained personnel responsible for flight control and ground control points (GCPs) for accurate georeferencing. The drone recorded 48 min of 4 K video at a horizontal flight speed of 21 metres per second across two consecutive flights. Aerial photography was performed with ground sample distances (GSDs) of 2, 5 and 10 mm, corresponding to flight heights of 2, 5 and 10 m, respectively. To achieve full coverage of the target area, the drone followed pre-set parallel transects spaced 2 m apart, surveying approximately 20 ha of mudflat. To mitigate the influence of environmental factors such as wind direction and lighting, multiple rounds of image acquisition were conducted until a complete dataset was obtained, and drone flights were conducted using an automated waypoint grid. Frames are extracted from videos instead of using static images because video streams can provide continuous image data, thereby increasing the number of samples in the dataset. Motion compensation is also performed between frames to reduce motion blur caused by the high-speed flight of the drone.

Super-resolution reconstruction

As shown in Fig. 2a, frames were extracted from a 4 K video at 4 Hz and cropped to 512 × 512 pixels to create HR images. Then, relevant LR images were generated in an image degradation model. LR is generated by HR through a bicubic downsampling operator $\mathscr{S}_s^{\downarrow}$, where the scale factor s is determined by the ratio of the target ground sampling distance GSD to that of HR. The paired HR and LR images were enhanced together and used as input to the SRR model. A series of SRR models was used for comparative experiments, and the hyperparameters were continuously optimized during the training process. Finally, the newly generated LR images were input to the trained SRR model, and the generated SR images were evaluated. This section will introduce the used SRR models, training configurations and evaluation metrics in detail.

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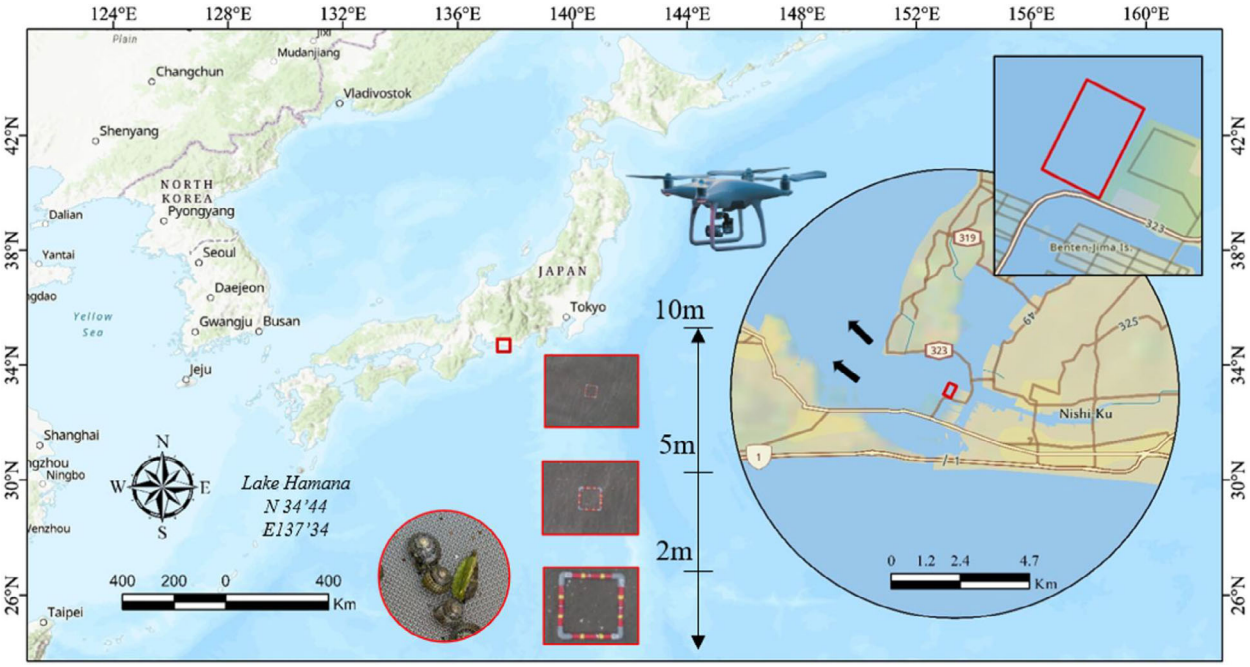

Figure 1. Drone-based field survey conducted at Lake Hamana, Hamamatsu City, Japan.

Network architectures

Deep learning-based SRR methods are mainly based on three basic architectures: CNN, residual network (ResNet) and generative adversarial network (GAN) (Anwar et al., 2020). CNN-based networks are usually simple in structure, fast in processing and focus on pixel features to determine interpolation. However, these algorithms often have problems with edge preservation and fine texture details, resulting in jagged edges and blocky artefacts. To address this problem, ResNet-based architectures combine information from both low-level and deep-level using shortcut connections, so that deeper layers can learn new features from the input data. However, when the image sampling rate (i.e., magnification) is too large, texture details may be lost. In this context, GAN-based methods can effectively fill in the missing details by simulating the degradation process of the real world.

In order to improve the resolution restoration effect, some specific SRR architectures have been designed (Fig. 3). The first SRR model, SRCNN, was implemented on only three convolutional layers to fit the nonlinear mapping relationship between low- and high-resolution images, achieving a higher peak signal-to-noise ratio (PSNR) (Shi et al., 2016). EDSR, which introduces a residual mechanism, improves the resolution by combining convolutional blocks and residual blocks (Hitawala et al., 2018). SRFBN uses feedback blocks for dense inter-layer jumps, which enhances the reconstruction effect. RCAN divides feature extraction into two parts, using convolutional layer heads to extract shallow features and RG structures to extract deep features. Finally, RDN uses densely connected convolutional layers to extract rich local features, forms continuous memories with residual dense blocks and optimizes feature utilization through 1×1 convolution to simplify data (Zhang et al., 2018).

Evaluation

In this study, to evaluate the Super-resolution reconstruction performance of the network, structural similarity (SSIM) and peak signal-to-noise ratio (PSNR) were used to evaluate the performance.

SSIM is a metric for subjective similarity assessment using brightness, contrast and structural properties to measure the structural similarity (0–1) between the generated image and the original image. A larger SSIM represents a high similarity between the SR image and the HR image. Luminance (L), Contrast (C) and Structure (S) are defined as follows:

$$L = \frac{2m_x m_y + c_1}{m_x^2 + m_y^2 + c_1} \tag{2-1}$$

$$C = \frac{2v_x v_y + c_2}{v_x^2 + v_y^2 + c_2} \tag{2-2}$$

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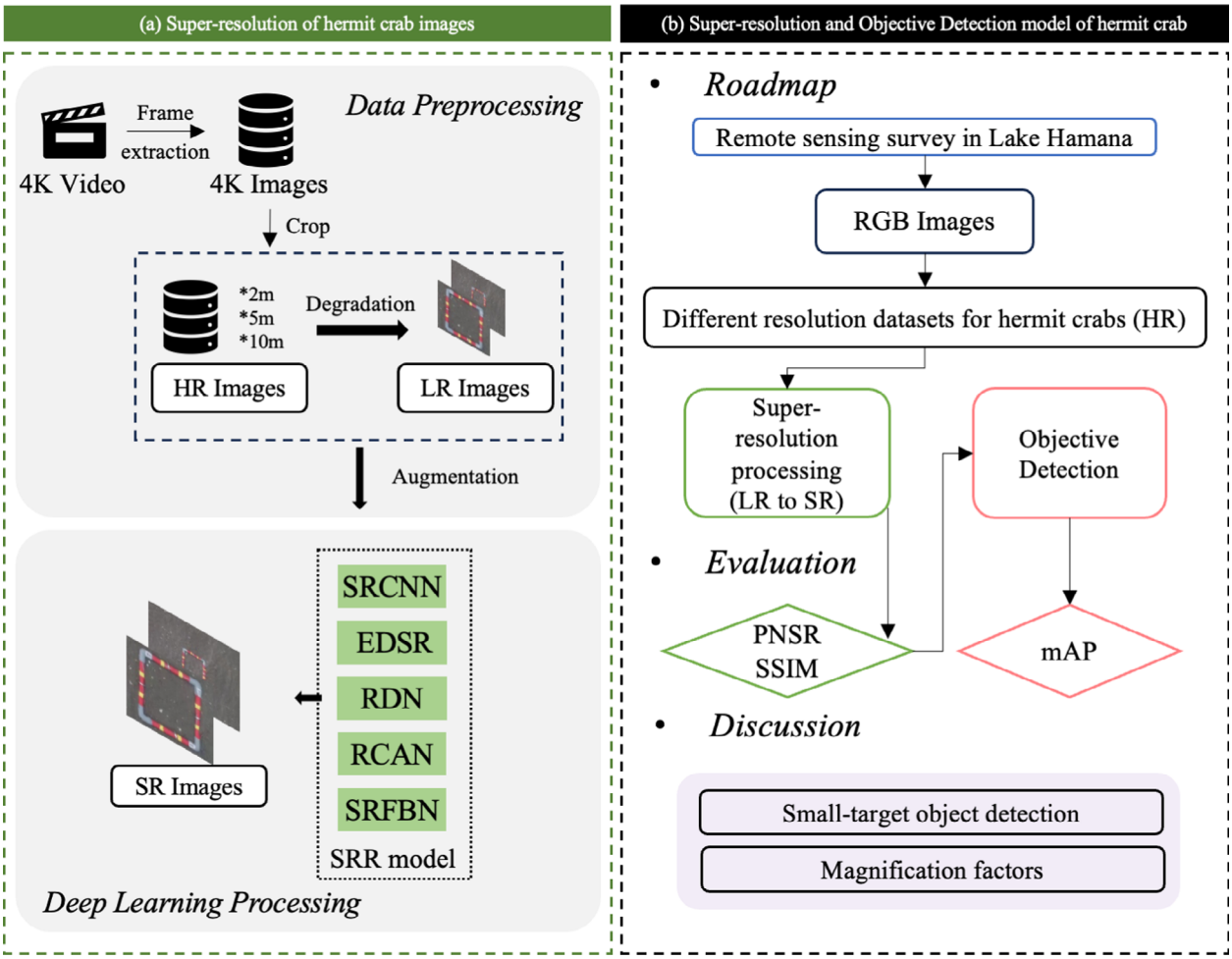

Figure 2. (a) Super-resolution processing of hermit crab images using deep learning models (SRCNN, EDSR, RDN, RCAN, SRFBN); (b) Overall system pipeline showing RGB image acquisition, resolution enhancement, object detection using V9-BENTHOS and evaluation metrics (PSNR, SSIM and mAP), with discussions focusing on small-object detection and magnification factor optimization.

$$S = \frac{v_{xy} + c_3}{v_x v_y + c_3} \tag{2-3}$$

where m_x is the mean of x; v_x is the variance of x; m_y is the mean of y; v_y is the variance of y; then the formula for SSIM is

$$\mathrm{SSIM}(x,y) = \frac{(2m_x m_y + c_1)(2v_{xy} + c_2)}{(m_x^2 + m_y^2 + c_1)(v_x^2 + v_y^2 + c_2)} \tag{2-4}$$

PSNR is a metric used to measure the quality of the generated image and its computational, which can be calculated by:

$$\mathrm{PSNR} = 10 \cdot lg_{10}\left(\frac{i^2_{\max}}{f_{\mathrm{MSE}}}\right) \tag{2-5}$$

$$f_{\mathrm{MSE}} = \frac{1}{\mathrm{AB}} \times \sum_{i=0}^{m-1}\sum_{i=0}^{n-1}[I(i,j) - K(i,j)]^2 \tag{2-6}$$

where $i_{\max}$ is the maximum value of the pixel value, and 1 is taken in this experiment; f_{MSE} is the mean square error. $[I(i,j) - K(i,j)]$, I and K are $A \times B$ monochrome images.

Hermit crab object detection

Network architectures

YOLOv9 is a key iteration of the YOLO series. By introducing innovative technologies such as the Programmable Gradient Information (PGI) module and the Generalized Efficient Layer Aggregation Network (GELAN), it

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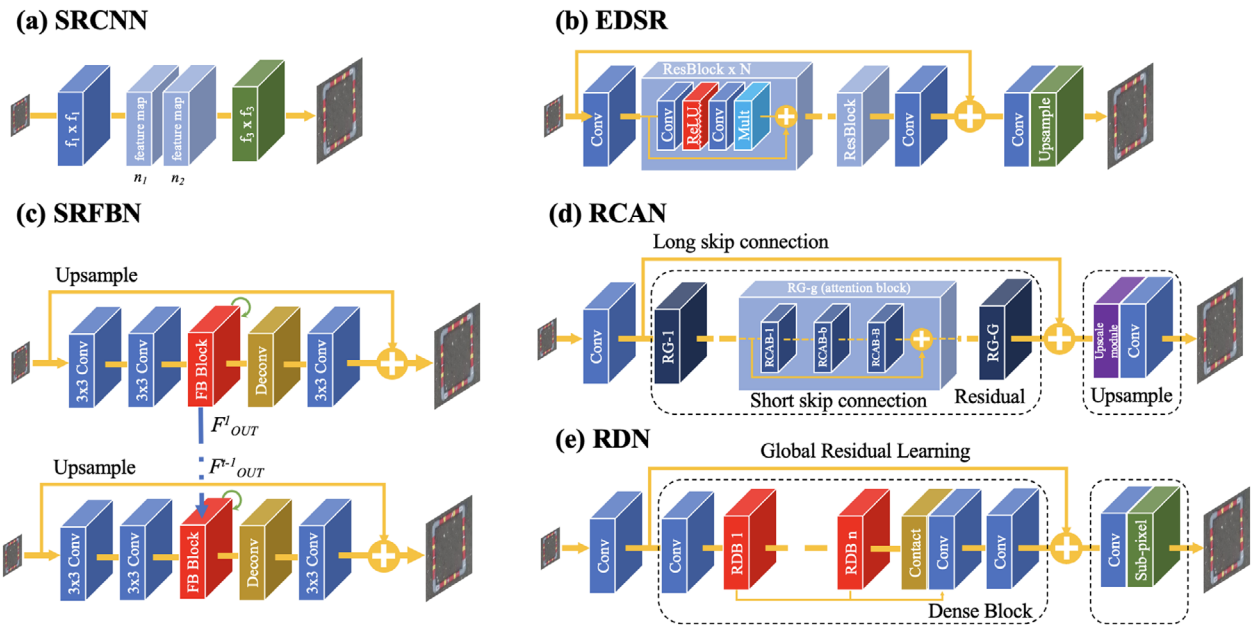

Figure 3. Architectures of Five SRR Networks Used in This Study. (a) SRCNN; (b) EDSR; (c) SRFBN; (d) RCAN; (e) RDN.

optimizes computational efficiency while maintaining or improving detection accuracy, demonstrating efficient object detection capabilities (Wang, Sun, et al., 2024; Wang, Zhao, et al., 2024). Its effectiveness has been verified on the MS COCO dataset. However, YOLOv9 is less used in detecting small objects such as hermit crabs. Since YOLOv9 divides the image into grids to predict multiple objects at the same time, small objects may span multiple grid cells, resulting in positioning errors. In addition, when the grid resolution is low, hermit crabs may only occupy a few pixels, resulting in unreliable detection. To address these limitations, this study proposes an improved algorithm, V9-BETHONS, which is specifically designed for small object detection, thereby improving the accuracy of hermit crabs (see Fig. 4).

Remote sensing images contain rich details and multi-scale features, demanding significant computational resources at high resolutions. Groupwise Separable Convolution (GSConv) optimizes computation and reduces parameters by decomposing standard convolution into multiple intra-group convolutions and pointwise convolutions. In hermit crab recognition, where the objects are small, GSConv mitigates overfitting and enhances object-background distinction.

Additionally, Bi-FPN introduces bidirectional feature fusion to improve interactions between hierarchical feature maps. Since hermit crabs are small objects with high detail sensitivity, they predominantly appear in low-level feature maps. Bi-FPN strengthens semantic information in high-level features while preserving spatial details in low-level features, leading to a more comprehensive feature representation.

In V9-BENTHOS, three lightweight modules are added to enhance multi-scale representation and effective receptive field, and mitigate small object loss. These modules include: (i) Bidirectional Weighted Feature Pyramid (Bi-GS-FPN), (ii) Asymmetric Convolution Enhancement (RepNCSPELAN4-AKConv) and (iii) Atrous Spatial Pyramid with Adaptive Downsampling (SPD_ADown). On 2 M datasets, mAP@50 is improved by 5.1% compared to the baseline YOLOv9, while the number of parameters increases from 50.7 M to 52.7 M (see Table 3). Training hyperparameters are detailed in the Supplementary Material.

Evaluation

Object detection for hermit crabs provides pixel-level information about the position of the hermit crab in the image and the strength of the recognition. In this study, three evaluation metrics commonly used in object detection were used to assess the accuracy of the obtained results: Precision (P), Recall (R) and mean Average Precision with intersection over unity (IoU) of 0.5 (mAP50). Each evaluation metric is defined as follows:

$$P = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{2-7}$$

$$R = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{2-8}$$

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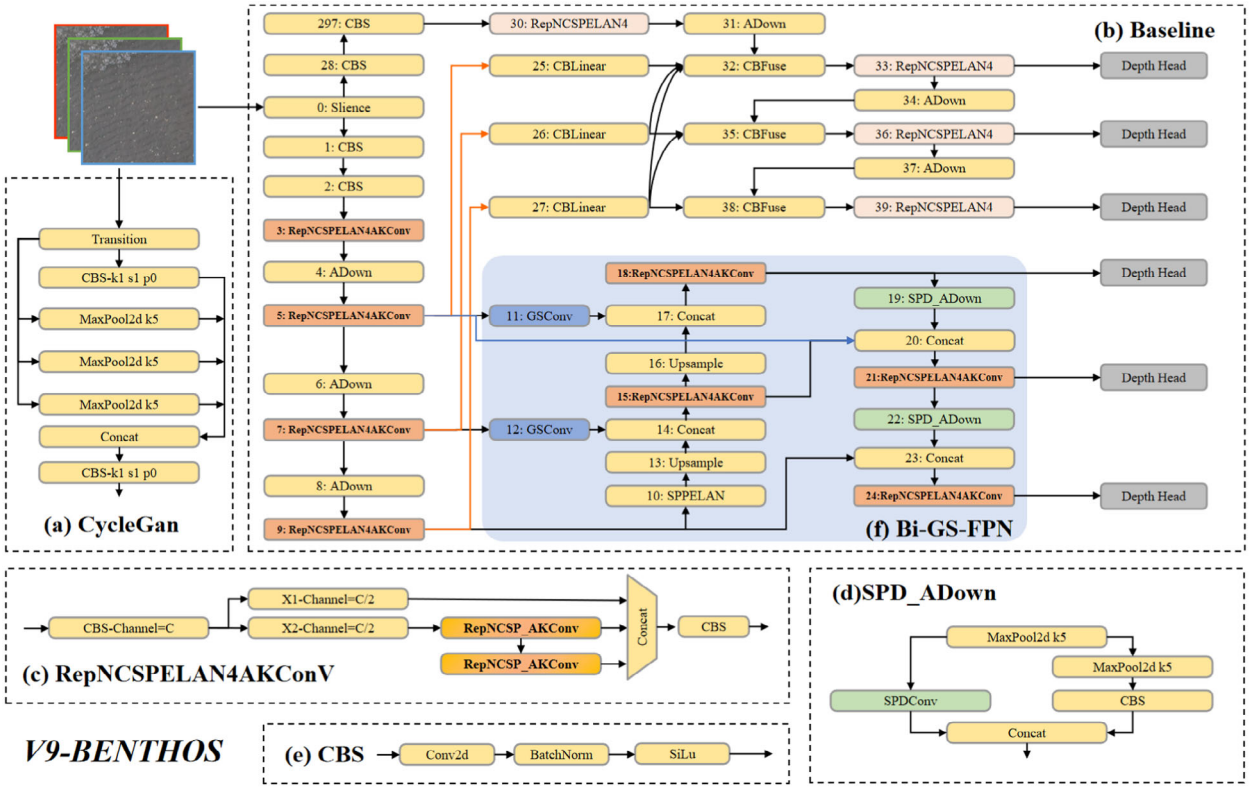

Figure 4. Schematic of the V9-BENTOS structure. (a) SPPELAN module for preliminary image sampling, (b) Baseline of V9-BENTHOS, (c) RepNCSPELAN4AKConv module for better feature extraction, (d) SPD-Adown downsampling module, (e) CBS convolution structure, (f) Bi-GS-FPN module.

$$\mathrm{AP} = \int_0^1 \mathrm{Precision}(r)dr \tag{2-9}$$

$$\mathrm{mAP50} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{AP}_i \tag{2-10}$$

where True Positive (TP) is a predicted frame with an IoU greater than or equal to 0.5 and the category is correct, False Positive (FP) is a predicted frame with an IoU less than or the category is incorrect, and False Negative (FN) is a true frame that was not predicted.

Results

Drone-based survey of hermit crabs

Experimental arrangement

The main objective of reconstructing hermit crab images using SRR is to obtain better SR images and achieve better performance in object detection processing. The whole training contains the following two parts:

Part 1: On the dataset from GSD $\approx [g_1]/[g_2]/[g_3]$ cm/pixel (nominate heights are 2, 5 and 10 m, respectively), images were input into five different SRR networks (SRCNN, SRFBN, EDSR, RCAN and RDN) for training and hyperparameter optimization. The fully trained models performed SRR on LR images to obtain SR images. The quality of SR images was evaluated using PSNR and SSIM to compare the performance of each algorithm.

Part 2: The original HR hermit crab dataset was first utilized to train the object detection network, based on which object detection was performed on the HR, Bicubic and SR test sets, and they were evaluated using precision, recall and mAP50. The facilitation of SR images for the object detection task is demonstrated.

SRR for benthos images

To cover mainstream SRR models, we selected SRCNN, EDSR, RDN, RCAN and SRFBN as representative baselines. As shown in Table 1, RDN achieved the highest or tied-highest PSNR/SSIM at 2 m/5 m/10 m and brought

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Table 1. Evaluation metrics for different super-resolution methods on the LR images (magnification factor × 4).

GSD	Metrics	Bicubic	SRCNN	SRFBN	EDSR	RCAN	RDN
2 m	PSNR (dB)	36.24	36.86	37.03	37.07	37.29	**37.32**
	SSIM (%)	83.94	85.61	85.89	85.99	86.78	**86.86**
5 m	PSNR (dB)	36.24	36.40	36.58	36.66	36.97	**37.04**
	SSIM (%)	83.94	85.13	85.45	85.59	86.44	**86.54**
10 m	PSNR (dB)	36.90	37.70	37.82	37.89	38.14	**38.15**
	SSIM (%)	86.09	87.71	87.94	88.03	88.50	**88.51**

Bold values indicate the best results among the compared methods for Table 1.

Table 2. Evaluation metrics of detection results of the 2 m images reconstructed by different SRR algorithms.

Metrics (%)	HR	Bicubic	SRCNN	SRFBN	EDSR	RCAN	RDN
Precision	96.3	84.3	84.8	91.3	92.7	92.6	**94.7**
Recall	94.1	75.1	79.0	76.6	76.1	84.5	**88.0**
mAP	97.2	83.8	85.3	88.8	89.5	92.4	**95.5**

Bold values indicate the best results among the compared methods for Table 2.

the largest mAP improvement in downstream detection (see Table 2). Given RDN's version stability and representativeness, this paper uses RDN as the primary SRR model. Section 4.3 briefly reviews the applicability and risks of recent methods.

As the number of epochs increases, the PSNR and SSIM gradually increase. When the loss tends to a minimum constant, the model converges. The HR image and the image reconstructed by the traditional interpolation method (Bicubic) are used for comparison. Table 1 shows that the PSNR and SSIM of the deep learning-based SRR method are larger than those of the Bicubic method, indicating that better reconstruction results are achieved.

From Table 1 and Fig. 5, SRCNN, as an early CNN-based super-resolution model, has a relatively simple structure that limits its ability to process complex textures and details, so it performs poorly in the SRR method. EDSR increases the network depth, so it has a significant improvement in processing high-frequency details of the image. The feedback mechanism of SRFBN updates the feature representation through multiple iterations, gradually improving the global quality of the super-resolution image. RCAN introduces a channel attention mechanism to adaptively adjust the weight of each channel, enhance the focus on important features and enable more balanced resolution improvements for different categories of objects in the image. RDN effectively improves feature reuse and information flow through the combination of Dense Connection and Residual Network and enhances the learning ability of the model, so that it not only optimizes global information but also pays more attention to the improvement of resolution in details and performs the best among the five tested SRR methods.

Object detection for SRR hermit crab images

A series of representative models was compared and evaluated under the same experimental conditions. For the centimetre-scale small-object detection and benthic

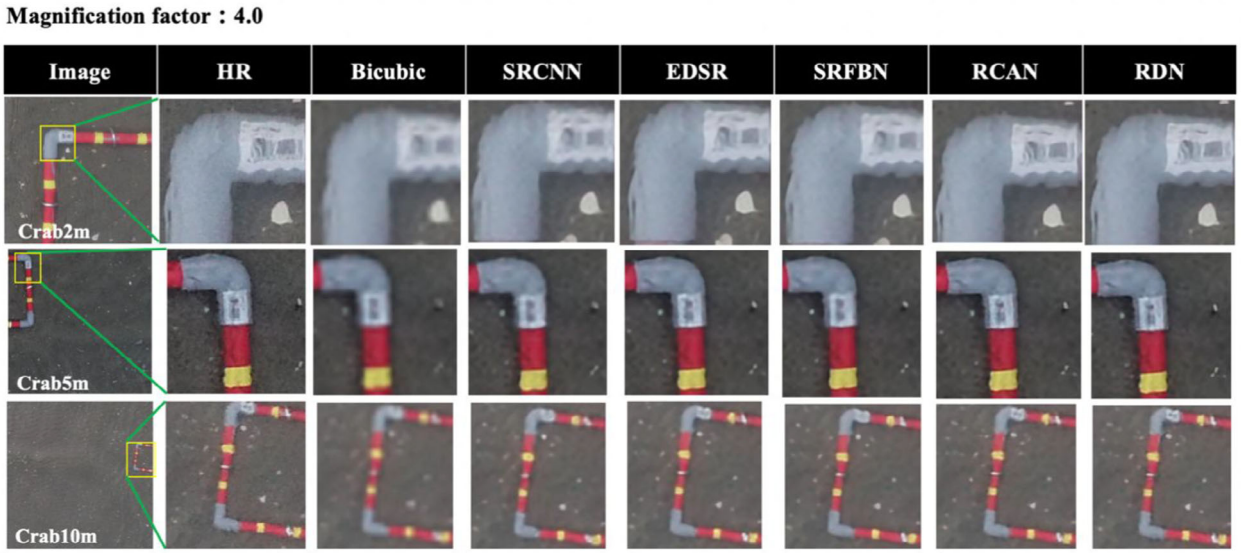

Figure 5. Comparison of visual effects of six super-resolution methods for reconstructing images.

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complex background tasks used in this study, further introducing later detectors (such as YOLOv10) under the same training paradigm would not significantly alter the SRR improvement on detection tasks. Instead, it would introduce additional variables, such as removing the NMS-free pipeline and impair reproducibility. Therefore, this study selected YOLOv9 and its improved version V9-BENTHOS qualitatively demonstrate the gains of SRR on detection tasks. They were therefore selected as the baselines in this paper, and V9-BENTHOS was constructed based on them. For further details, please refer to Chapter 4.1.

The SR hermit crab dataset is used to perform object detection tests on the V9-BENTHOS network proposed in this paper to verify the optimization effect of the SRR method on the resolution improvement of object details in the object detection task.

During the training process, the IoU starts to increase dramatically and finally converges to about 0.5. At this point, the object detection effect is evaluated in terms of three metrics: precision, recall and mAP@50. Table 2 shows the detection results on HR images, Bicubic images and corresponding SR images generated by the five SRR networks. Among them, the highest detection accuracy (96.3%), recall (94.1%) and mAP (97.2%) are obtained on HR images. This is due to the fact that HR is directly captured by the HD device, which has more texture features compared to the high resolution obtained from the low-resolution SRR processing. The detection metric values of the different SR images are very close to each other, and all of them are substantially ahead of the Bicubic images. The RDN network with the best precision, recall and mAP on the SR test set improved by up to 12.3, 17.1 and 14.0% compared to the results on the Bicubic test set.

In order to intuitively show the differences between the hermit crab images reconstructed by different algorithms (Fig. 6) shows the box selection results and confidence values of different SRR algorithms in the object detection task. Some hermit crab objects that are difficult to detect in the Bicubic method are successfully recognized in the SRR method. Notably, confidence scores rising above 0.5 indicate a qualitative shift in object detection, as this threshold is commonly used to accept detections as valid. As shown in Fig. 6, applying the SRR method results in a qualitative shift in the confidence scores for hermit crab detection. For example, in both RDN and SRCNN, the confidence score for the first crab increases from 0.49 (bicubic) to over 0.60; similar results are observed across multiple SRR models. These improvements demonstrate that SRR enables the detection network to achieve higher confidence scores, validating its effectiveness in improving recognition performance. In addition, the confidence value of the same object is greatly improved after SRR processing.

Ablation experiment

To further evaluate the effectiveness of the proposed feature extraction module in enhancing the receptive field and learning multi-scale representations, segmentation experiments were conducted on four network variants without additional modules.

The same object detection evaluation metrics were used to evaluate the four improved combined networks. As shown in Table 3, the original YOLOv9 model achieves

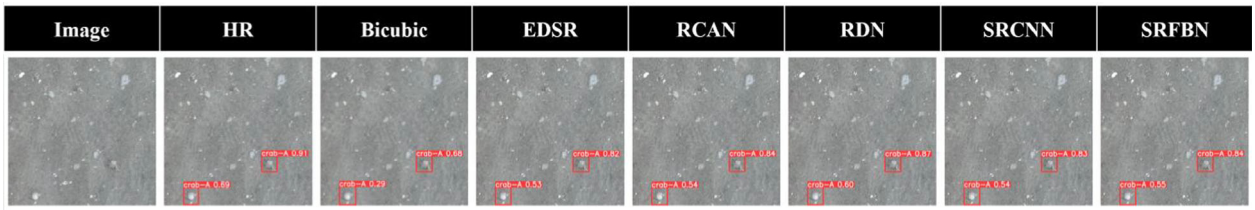

Figure 6. Effects of different SRR treatments on recognition in V9-BENTHOS.

Table 3. Detection results of different network improvement combinations.

Model	RepNCSPELAN4AKConv	Bi-GS-FPN	SPD_ADown	P (%)	R (%)	mAP@50 (%)	Params.
YOLOv9				90.3	86.5	90.4	50.7 M
YOLOv9-1	√			92.9	92.0	94.3	50.1 M
YOLOv9-1-2	√	√		93.5	93.3	95.6	50.5 M
YOLOv9-1-2-3 (V9-BENTHOS)	√	√	√	96.3	94.1	97.2	52.7 M

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relatively good detection results, while the V9-BENTHOS model is further improved on its basis. The results show that although the objects of hermit crabs in remote sensing images are small and have fewer input features, the three feature extraction modules proposed in this paper enable the network to extract features from different receptive fields, which increases the width of the network, enriches the information of the network and improves the performance of the network without taking up a lot of space.

Specifically, the RepNCSPELAN4AKConv module alone significantly improves precision and recall. The addition of the Bi-GS-FPN module further enhances multi-scale feature fusion, resulting in improved recall and mAP. The SPD_ADown module has the highest overall performance, indicating that it is very effective in improving feature utilization and spatial information retention. It is noteworthy that these improvements were achieved with a moderate increase in model parameters (from 50.7 M to 52.7 M), which demonstrates the efficiency of the method. This means that using V9-BENTHOS in real ecological monitoring can detect more real individuals from the background (e.g. sand grains and blisters), and the estimation of the distribution is closer to the real ecological state.

Table 4. The detection results of various networks on different 2 m dataset.

	mAP@50 (%)			Computational cost	
Models	HR	Bicubic	SR	Params. (million)	Size (MB)
Faster R-CNN	90.6	78.2	85.5	28.4	113.5
SSD	88.2	77.9	78.3	26.3	95.5
YOLOv3tiny	89.7	48.1	65.5	8.7	17.4
YOLOv4	86.1	68.5	85.0	64.4	256.3
YOLOv5s	86.7	62.9	83.4	7.0	14.5
YOLOv5Lite-g	86.6	16.7	44.9	5.5	11.3
YOLOX-nano	86.5	70.0	81.6	0.9	3.9
YOLOv7tiny	53.7	32.8	45.3	6.0	12.3
YOLOv8n	71.0	56.8	63.9	3.0	6.2
YOLOv9	90.4	79.4	90.4	50.7	102.8
V9-BENTHOS	97.2	83.8	95.5	52.7	106.9

Discussion

In the previous sections, based on the self-built Lake Hamana hermit crab dataset, an efficient method combining a super-resolution network and an improved object detection algorithm-V9 BENTHOS, for hermit crab survey was proposed. In this section, related issues of super-resolution and object detection algorithms in practical engineering applications will be discussed.

Small-object detection algorithms on the SR image

To further illustrate the advanced nature of V9-BENTHOS, 10 different models for common benthic organism detection were trained in Discussion 1, including two CNN-based models and representative networks from YOLOv3 to v9. Each variant name represents a different version and structure within the series, and identification is performed on a self-built hermit crab dataset, and the mAP results and computational cost are compared, as shown in Table 4.

Compared with other networks, the proposed V9-BENTHOS and its baseline model demonstrate a significant improvement in recognition accuracy while maintaining a relatively low computational cost. V9-BENTHOS achieves the highest mAP score among the 11 evaluated networks, highlighting the effectiveness of its multi-layer feature extraction and fusion mechanism. An analysis of the experimental results reveals that Faster R-CNN (Ren et al., 2016), which builds upon CNN by introducing shared convolutional layers between the feature extractor and detection head, performs well in hermit crab detection. Its hierarchical architecture enables high accuracy but comes with the trade-off of increased model size.

Another primary challenge in benthic organism recognition lies in differentiating complex foreground-background relationships. Recent studies have addressed this by incorporating lightweight classifiers such as ShuffleNet to perform scene classification prior to segmentation, thereby reducing the computational burden (Bi et al., 2024). Single-stage detectors such as SSD (Hussain et al., 2024), while efficient and well-suited for real-time detection of larger objects, for example, deep-sea fish, encounter limitations when applied to millimetre-scale benthic organisms like hermit crabs. As a lightweight variant in the YOLO family, YOLOv3-tiny has demonstrated rapid detection capabilities for small objects and is widely deployed in underwater vehicles for efficient marine surveys (Mehta & Ozturk, 2018). Subsequent iterations such as YOLOv4 have integrated components including attention mechanisms, dense connections and enhanced feature interfaces to improve adaptability in underwater scenarios (Zhao et al., 2022). However, these additions result in a more complex and computationally heavy network (Zhao et al., 2024). From YOLOv5 to YOLOv13n, research has increasingly focused on ultra-lightweight architectures suitable for deployment on mobile platforms in marine environment (Liu et al., 2026; Wang, Sun, et al., 2024; Wang, Zhao, et al., 2024; Zhao et al., 2024, 2026). These models have been applied to the detection

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of sea cucumbers, starfish, shellfish and other benthos species. While these lightweight networks achieve moderate recognition accuracy with minimal computational overhead, their simplified structures often lack robustness in complex benthos imaging conditions. For instance, under downsampled inputs such as Bicubic interpolation, recognition accuracy for small hermit crab objects becomes inconsistent and less reliable (Ma et al., 2024; Zhao et al., 2024).

Effect of SRR magnification factor on hermit crab detection

In SRR processing, the magnification factor refers to the fixed upsampling ratio used to enlarge LR images to match the dimensions of HR images (Zhao et al., 2024). Determining the optimal magnification factor is essential in practical engineering applications, as it enables a balance between computational efficiency and enhancement performance (Zhao et al., 2024). In this study, experiments were conducted using drone-captured images at GSD $\approx [g_1]$ cm/pixel (2 m) and GSD $\approx [g_2]$ cm/pixel (5 m). The HR image resolution was fixed at 320×320 pixels, while the corresponding LR images were obtained through $\times 4$ downsampling, resulting in a resolution of 80×80 pixels. These LR images were randomly split into training (70%), validation (10%) and testing (20%). The magnification factors ranged from $\times 1$ to $\times 5$, where $\times 1$ corresponds to the original LR images collected under real-world conditions. The RDN identified in Section 3.1.2 as the most effective SRR model, was employed to generate enhanced images at $\times 2$ (160×1600), $\times 3$ (240×240), $\times 4$(320×320) and $\times 5$ (400×400) resolutions. These super-resolved outputs were subsequently used as new test inputs in the proposed V9-BENTHOS detection framework, allowing for evaluation of object detection performance and recognition confidence under varying magnification conditions.

Figure 7 illustrates the object detection performance of the V9-BENTHOS model on SR images generated with different magnification factors, highlighting both object localization and confidence scores. The confidence score represents the model's estimated probability of the presence of a hermit crab, ranging from 0 to 1. At a shooting height of 2 m, where the object occupies a relatively large portion of the image, the detection confidence remains consistently high (> 50%) across the HR and all SR images ($\times 1$ to $\times 5$). In contrast, at a height of 5 m, the hermit crab appears significantly smaller, resulting in lower detection confidence for both HR and LR inputs. However, application of SRR with magnification factors $\times 2$ to $\times 5$ substantially enhances recognition confidence. Overall, higher magnification levels improve detection performance by providing finer image details. Nevertheless, the performance gains between $\times 4$ and $\times 5$ are marginal, despite the increased image resolution, while the computational cost at $\times 5$ is notably higher. Therefore, a magnification factor of $\times 4$ offers an optimal trade-off between detection accuracy and computational efficiency. This finding aligns with previous studies on the selection of magnification levels in SR-based detection tasks (Zhao et al., 2024).

Ecological implications and applications

As demonstrated before, transformer-based and real-degradation SRR (such as SwinIR and Real-ESRGAN) have potential in natural scene reconstruction. However, in the low-texture or noisy intertidal mudflat images used in this study, they exhibit a tendency to hallucinate details, potentially amplifying spurious features and misleading small-object detection. This suggests that future

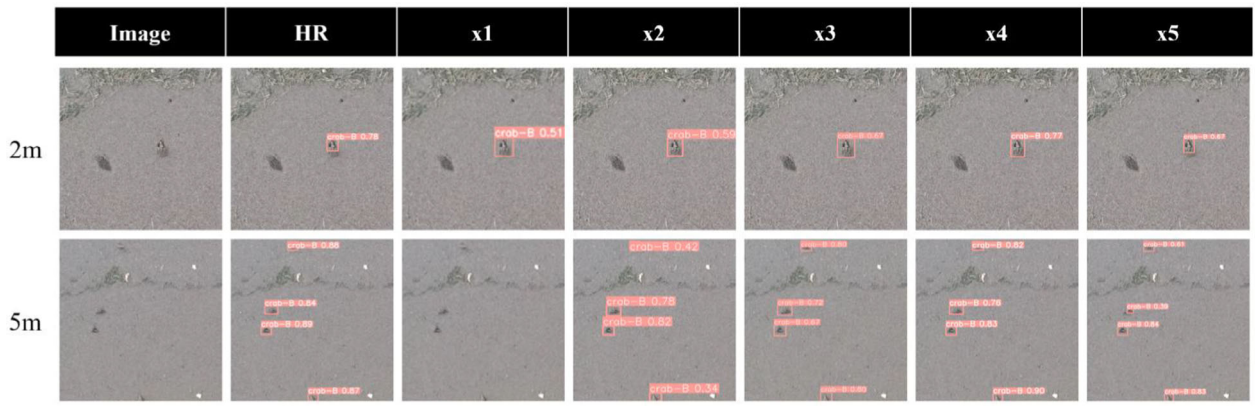

Figure 7. Effects of different magnification factors on detection in V9-BENTHOS.

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work should adopt task-aware training and evaluation, rather than focusing solely on reconstruction metrics like PSNR and SSIM.

The method proposed in this study has important implications for practical ecological applications. It enables efficient and non-destructive acquisition of high-resolution quadrat data, which can be used to assess hermit crab population density and spatial distribution patterns. By analysing high-resolution images collected at a height of 2 metres, we preliminarily estimated the average density of hermit crabs in the area to be approximately 1.2 crabs per square metre. As shown in Fig. 8, the spatial distribution of hermit crabs was heterogeneous, with pronounced clustering across the mudflat; the illustrated quadrat was placed within a locally aggregated area, while surrounding regions exhibited much lower densities, combined with V9-BENTHOS detection results, this clustered distribution may be related to the availability of empty snail shells, substrate type and foraging activity. This clumping pattern can affect detection performance, but our proposed V9-BENTHOS algorithm, integrating SRR with a modified YOLOv9, effectively identifies individuals within these clumped clusters by enhancing image detail and improving small object detection, thereby improving the accuracy of population estimates. This research has applications not only in ecological monitoring, such as assessing the population status of *Pagurus minutus*, but also in aquaculture management. For instance, drone images can be used to quantify individual size, guiding harvest planning and market timing to maximize farmers' profits.

Conclusion

This study presents an integrated approach for efficient benthic organism detection by combining deep learning-based SRR with an improved YOLOv9 framework, named V9-BENTHOS. Field experiments objecting hermit crabs at Lake Hamana demonstrated the effectiveness of this method under real-world environmental constraints. Among several SRR models evaluated, the RDN provided the highest image reconstruction quality, significantly enhancing detection performance in downstream tasks. The proposed V9-BENTHOS model achieved superior accuracy (mAP@50 of 97.2%) with minimal increases in computational complexity. Ablation studies confirmed the contribution of each module to multi-scale feature extraction and receptive field expansion. Furthermore,

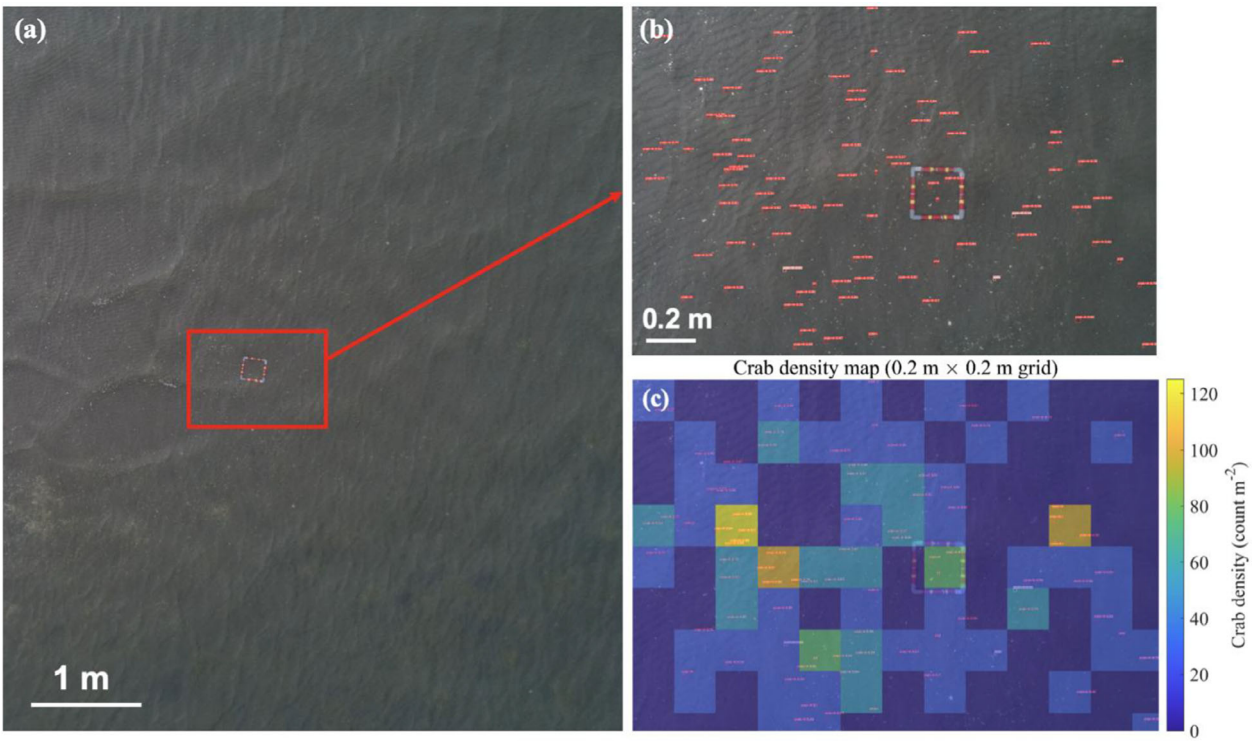

Figure 8. Spatial distribution and density mapping of hermit crabs: (a) overview of a sample survey area with high abundance of crabs; (b) high-resolution image of the quadrat (0.2 × 0.2 m grid), overlaid with detection results; (c) hermit crab density map derived from detection results, illustrating pronounced spatial heterogeneity and localized clustering.

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comparative experiments with ten existing detection algorithms established the superiority of V9-BENTHOS in both recognition accuracy and model efficiency, particularly in detecting small-scale benthic organisms. The analysis of different magnification factors also highlighted × 4 as an optimal balance between image detail and computational cost. Overall, the proposed methodology demonstrates a robust, scalable and cost-effective solution for ecological monitoring in shallow-water environments. It holds great potential for broader application in the detection of small marine organisms using low-cost UAV platforms and standard imaging equipment, thereby supporting long-term biodiversity assessment and conservation efforts.

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Author Contributions

Fan Zhao: Conceptualization; methodology; software; data curation; validation; formal analysis; investigation; funding acquisition; writing – original draft; writing – review and editing; project administration; resources. **Bangzhang Ma:** Conceptualization; methodology; investigation; formal analysis; writing – original draft. **Dianhan Xi:** Conceptualization; methodology; software; visualization; writing – review and editing. **Jiaqi Wang:** Methodology; software; validation; formal analysis; visualization; writing – review and editing. **Yijia Chen:** Conceptualization; data curation; investigation; project administration; writing – review and editing. **Yongying Liu:** Methodology; validation; visualization; project administration. **Xinlei Shao:** Methodology; validation; visualization; project administration. **Mowen Zhang:** Data curation; formal analysis; visualization; project administration. **Guocheng Zhang:** Software; validation; writing – review and editing; resources. **Jundong Chen:** Methodology; supervision; writing – review and editing. **Katsunori Mizuno:** Investigation; validation; supervision; resources; data curation; writing – review and editing.

Data Availability Statement

The datasets from this study will be made available from the corresponding author on reasonable request.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1.

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